MILITARY STANDARD

MIL-STD-1320-53C
(NAVY)
23 October 1990

TRUCKLOADING

UNIT LOAD FOR UNDERWAY REPLENISHMENT

BOMB, GENERAL PURPOSE, MK 84 AND MODS

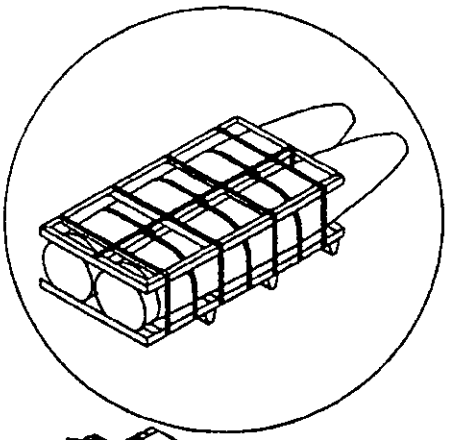
(2000 LB)

IN ADAPTER, UNIT LOAD, MK 79 MOD 0

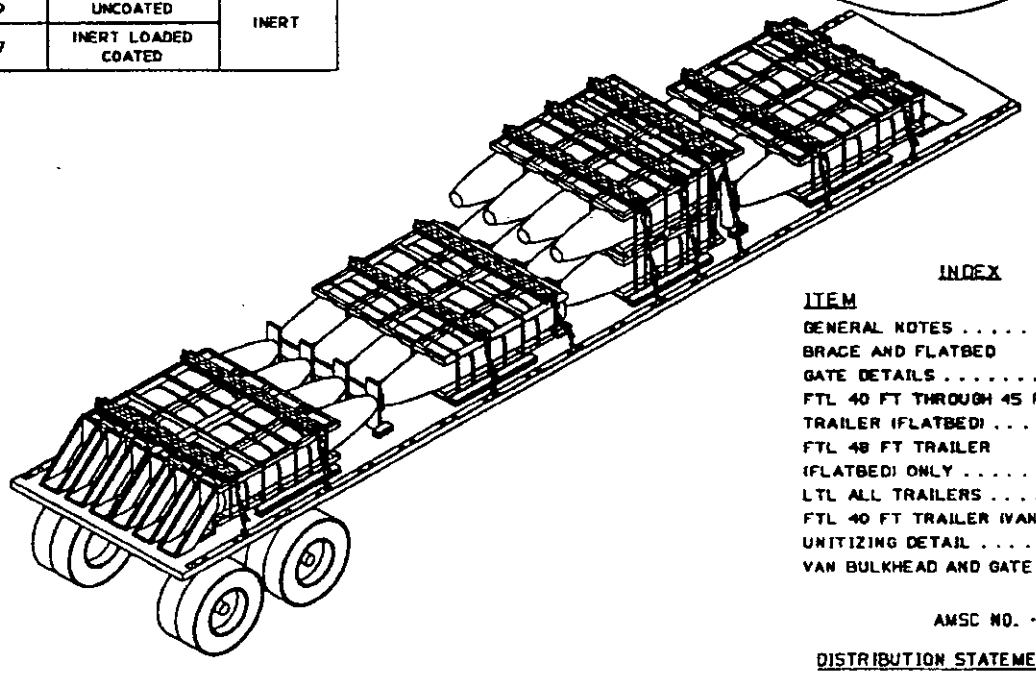
SUPERSEDING
MIL-STD-1320-53B
10 JANUARY 1977

UNIT LOAD DATA	UNCOATED	COATED
UNIT LOAD DOCUMENT	MIL-STD-1323-127	MIL-STD-1323-253
DIMENSIONS	99 L X 38 W X 24 1/4 H	99 L X 38 W X 24 5/8 H
GROSS WEIGHT	TRITONAL 4077 LBS OR H6 4111 LBS Δ	4124 LBS Δ
CUBE	52.8 CU FT	53.6 CU FT

Δ DO NOT USE FOR SHIPPING WEIGHT



HAZARD CLASSIFICATION		
DODIC/NALC	ITEM NAME	DOT
F126/F127/F128 F270/F275/F281	EXPLOSIVE LOADED UNCOATED	CLASS A
F272/F274/F278	EXPLOSIVE LOADED COATED	
F262/F268/F269 F273/F279	INERT LOADED UNCOATED	INERT
F264/F277	INERT LOADED COATED	



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AMSC NO. - N/A

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REV LTR	REVISION DESCRIPTION	DATE	TDA	SYSCOM
C	REDUCED TRUCKLOAD WEIGHTS AND UPDATED DOCUMENT.	16/25/4		
			APPROVAL	

NOTES:
1. UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED
FOR HIGHWAY SHIPMENT ONLY.

S/ M. S. GRAY 1/6/77
SIGNATURE TDA, WPNSTA EARLE DATE

S/ J. E. KELLY 1/6/77
SIGNATURE AIRSYSCOM BY DIRECTION DATE

ORIGINATOR S/ CHARLES MC BRIDE 1/6/77
DATE

WPNSTA EARLE, NEW JERSEY

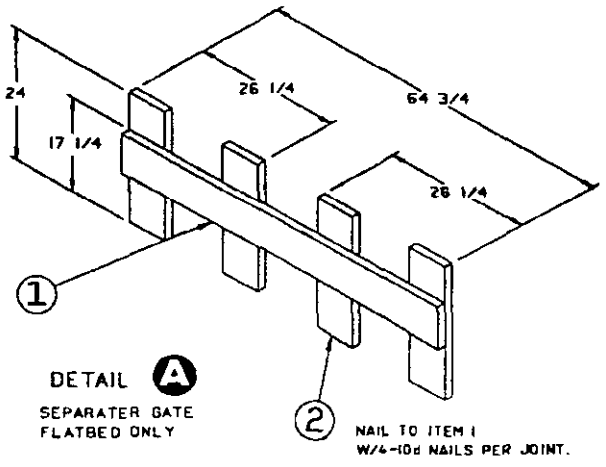
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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING UNIT LOADS FOR UNDERWAY REPLENISHMENT OF MK 84 (2000 LB) BOMBS WHEN THE BOMBS ARE PALLETIZED IN ACCORDANCE WITH MIL-STD-1323-127 (NAVY) OR MIL-STD-1323-253 (NAVY).
2. THE PROCEDURES DESCRIBED HEREIN ARE INTENDED FOR 40 FT THROUGH 48 FT FLATBED TRAILERS WITH THE TRAILERS TANDEM AXLES LOCATED IN THE "WESTERN POSITION" (AT THE EXTREME REAR OF THE TRAILER). DO NOT USE TRAILERS WITH METAL FLOORS.

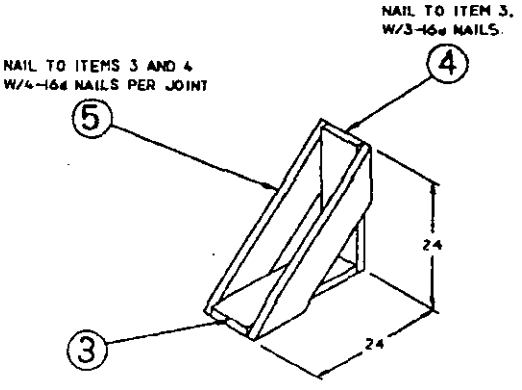
WARNING

3. FLATBED TRAILERS ARE THE AUTHORIZED MODE OF TRANSPORTATION. VANS ARE AUTHORIZED ONLY WHEN FLATBED TRAILERS ARE NOT AVAILABLE AND SHIPMENT MUST BE MADE BECAUSE OF MILITARY NECESSITY.
4. POSITION NOSE END OF BOMBS IN DIRECTION SHOWN TO PREVENT AXLE OVERLOAD.
5. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
6. CHAINS AND STEEL STRAPPING MAY BE USED INTERCHANGEABLY ON A ONE TO ONE BASIS FOR TIE DOWNS.
7. STEEL STRAPPING USED AS TIE DOWNS SHALL BE 2 X .050 AND SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).
8. CHAINS, FITTINGS AND LOAD BINDERS SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY) EXCEPT THAT THE CHAIN/GRAB HOOKS SHALL BE ATTACHED TO THE STAKE POCKETS (NOT AROUND THE RUB RAIL).
9. PROTECTOR BOARDS (DOUBLED 2 X 6'S) SHALL BE USED UNDER THE TIE DOWNS . WHEN USING STEEL STRAPPING, THE STRAPS SHALL BE STAPLED TO THE PROTECTOR BOARD USING 2 INCH STRAPPING STAPLES. WHEN USING CHAIN, THE CHAINS SHALL BE SECURED TO THE PROTECTOR BOARD BY DRIVING A 10d NAIL THROUGH THE CHAIN LINK AND CLINCHING THE NAIL OVER THE CHAIN. FIVE STAPLES/NAILS SHALL BE USED FOR EACH TIE DOWN.
10. PRIOR TO LOADING THE TRAILER AND DURING PRELOADING INSPECTION, REQUIRED BY NAVSEA OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS AND LOAD BINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS, FITTINGS AND LOAD BINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOAD BINDER.
11. UNLESS OTHERWISE SPECIFIED, DUNNAGE LUMBER, STEEL STRAPPING, STRAP SEALS AND NAILS SHALL BE AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).
12. AFTER THE BLOCKING, BRACING AND TIE DOWNS HAVE BEEN INSPECTED, THE CONTAINERS SHALL BE COMPLETELY COVERED WITH FIRE RESISTANT WATERPROOF TARPAULINS. THE TARPAULINS MAY BE UNDER THE TIE DOWNS.
13. AFTER THE TARPAULINS ARE IN PLACE, ATTACH "EXPLOSIVES A" PLACARDS TO BOTH SIDES, FRONT AND BACK OF THE TRAILER (PLACARDS NOT REQUIRED FOR INERT LOADED BOMBS) AND ATTACH THE SHIPPING DOCUMENTS TO AN ACCESSABLE AREA ON THE BACK DECK OF THE TRAILER.
14. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).



DETAIL A
SEPARATER GATE
FLATBED ONLY

NAIL TO ITEM 1
W/6-10d NAILS PER JOINT.



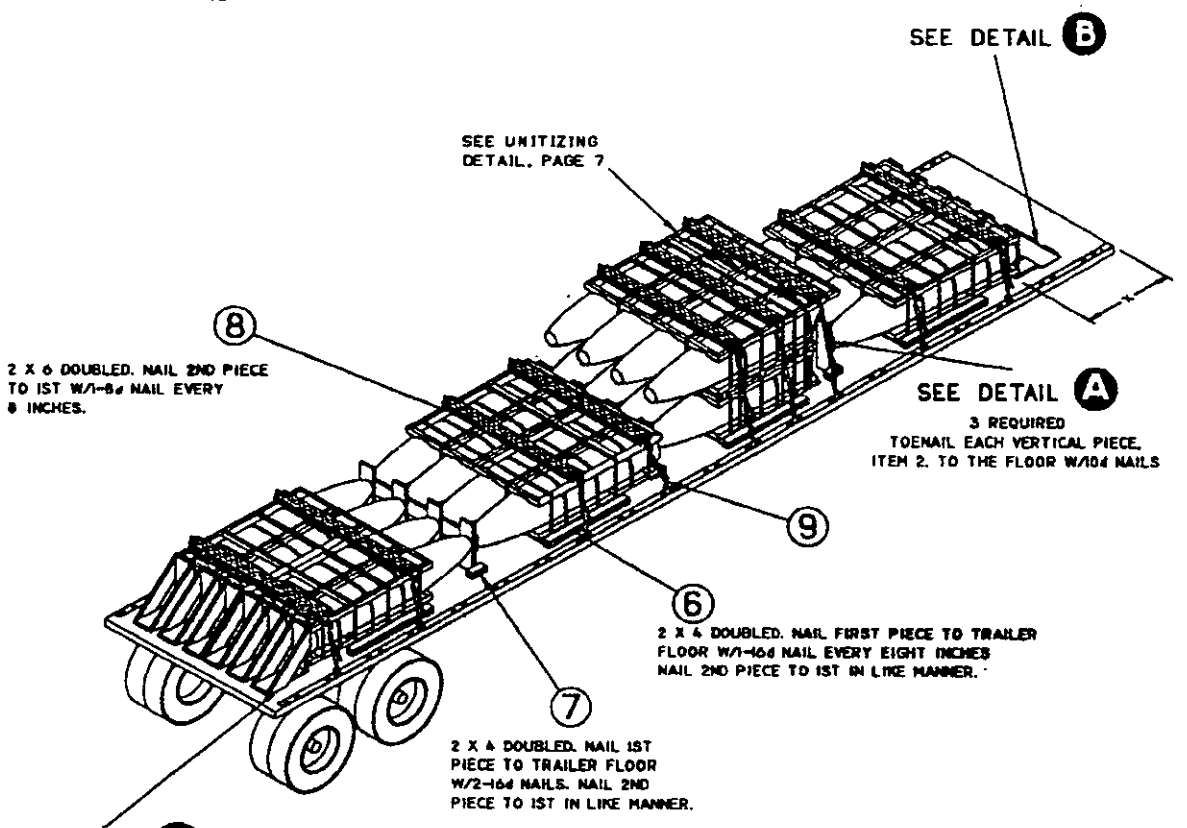
DETAIL B
BRACE
FLATBED AND VAN

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FTL 40 FT THROUGH 45 FT TRAILER (FLATBED)

NOTE

TARPAULINS OVER CONTAINERS OMITTED FOR CLARITY.



TRUCKLOAD DATA

NUMBER OF UNIT LOADS 10
WEIGHT . . . UNCOATED TRITONAL 40770 LBS Δ
WEIGHT . . . . . UNCOATED HG 41110 LBS Δ
WEIGHT . . . . . COATED 41240 LBS Δ

Δ DO NOT USE FOR SHIPPING WEIGHT

TRAILER LENGTH	DIMENSION X
40 FT	45
42 FT	55
44 FT	65
45 FT	70

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
9	9	CHAIN AND LOAD BINDER	SEE NOTE 8 ON PAGE 2	5/16 OR 3/8
16	8	PROTECTOR BOARD	SEE NOTE 11 ON PAGE 2	2 X 6 X 78
12	7	GATE CLEAT	SEE NOTE 11 ON PAGE 2	2 X 4 X 8
16	6	SIDE CLEAT	SEE NOTE 11 ON PAGE 2	2 X 4 X 46
24	5	BRACE, DIAGONAL	SEE NOTE 11 ON PAGE 2	2 X 8 X TO SUIT
12	4	BRACE, UPRIGHT	SEE NOTE 11 ON PAGE 2	2 X 8 X 24
12	3	BRACE, CLEAT	SEE NOTE 11 ON PAGE 2	2 X 8 X 22 1/2
12	2	VERTICAL	SEE NOTE 11 ON PAGE 2	2 X 8 X 24
3	1	HORIZONTAL	SEE NOTE 11 ON PAGE 2	2 X 8 X 64 3/4

LIST OF MATERIALS

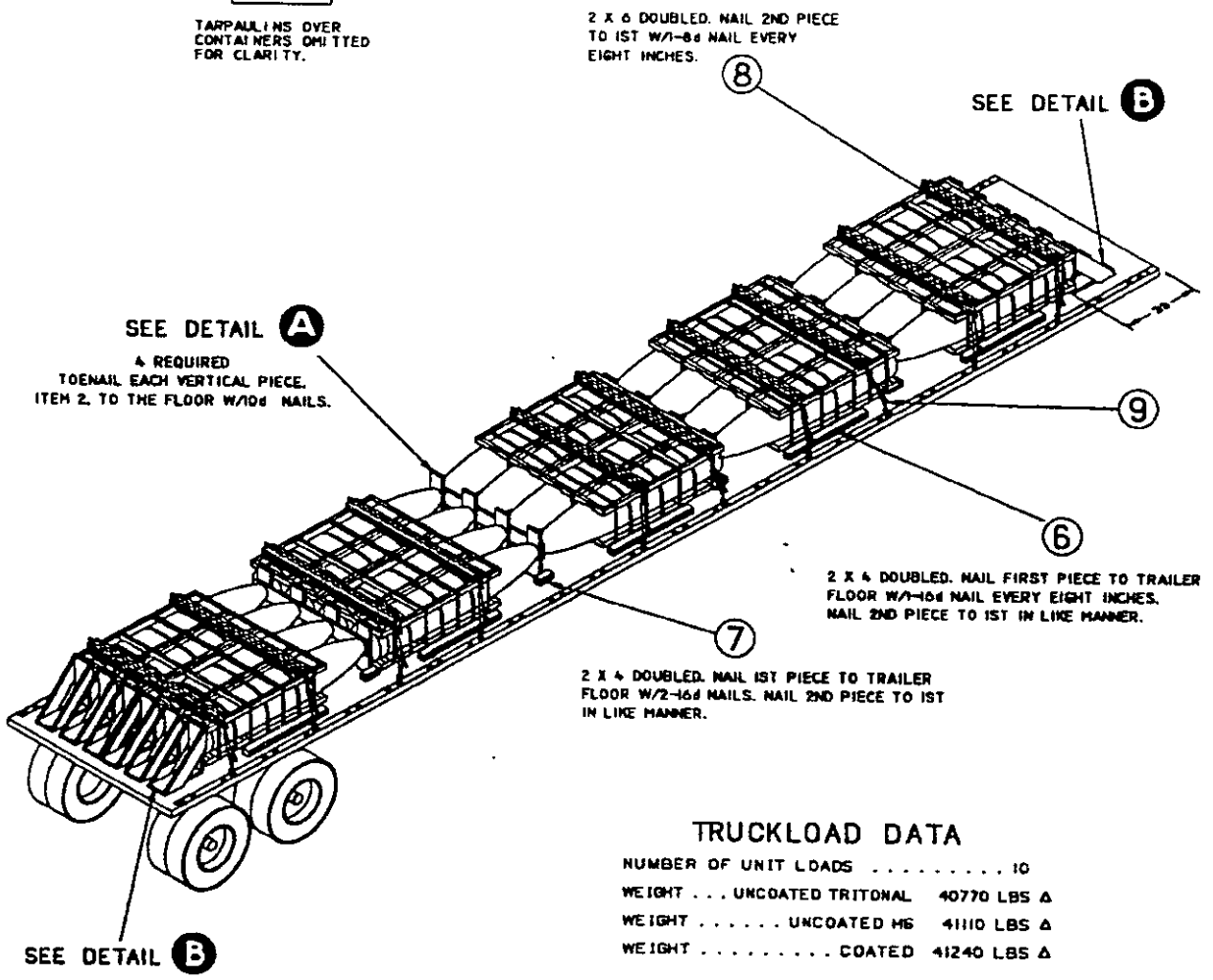
FTL 40 FT THROUGH 45 FT TRAILER (FLATBED)
(40 FT TRAILER SHOWN)

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FTL 48 FT TRAILER (FLATBED) ONLY

NOTE

TARPAULINS OVER CONTAINERS OMITTED FOR CLARITY.



TRUCKLOAD DATA

NUMBER OF UNIT LOADS 10
WEIGHT . . . UNCOATED TRITONAL 40770 LBS Δ
WEIGHT . . . . . UNCOATED HS 41110 LBS Δ
WEIGHT . . . . . COATED 41240 LBS Δ

Δ DO NOT USE FOR SHIPPING WEIGHT

LIST OF MATERIALS

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
10	9	CHAIN AND LOAD BINDER	SEE NOTE 8 ON PAGE 2	5/16 OR 3/8
20	8	PROTECTOR BOARD	SEE NOTE 9 & 11 ON PAGE 2	2 X 6 X 78
16	7	GATE CLEAT	SEE NOTE 11 ON PAGE 2	2 X 4 X 8
20	6	SIDE CLEAT	SEE NOTE 11 ON PAGE 2	2 X 4 X 46
24	5	BRACE, DIAGONAL	SEE NOTE 11 ON PAGE 2	2 X 8 X TO SUIT
12	4	BRACE, UPRIGHT	SEE NOTE 11 ON PAGE 2	2 X 8 X 24
12	3	BRACE, CLEAT	SEE NOTE 11 ON PAGE 2	2 X 8 X 22 1/2
16	2	VERTICAL	SEE NOTE 11 ON PAGE 2	2 X 8 X 24
4	1	HORIZONTAL	SEE NOTE 11 ON PAGE 2	2 X 8 X 64 3/4

FTL 48 FT TRAILER (FLATBED) ONLY

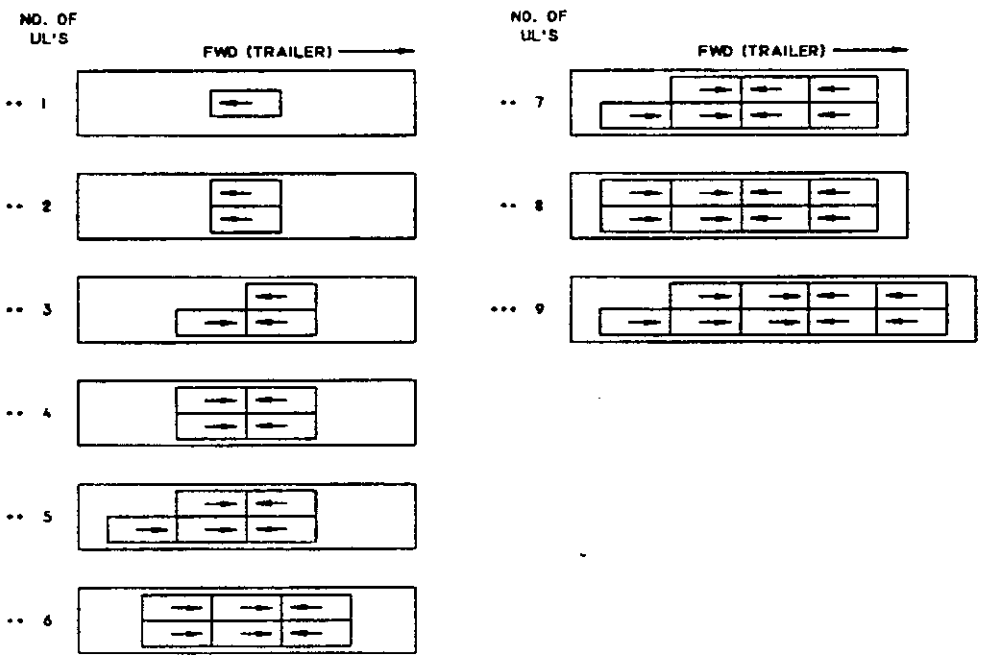
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LTL 40 FT THROUGH 48 FT TRAILERS

1. WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD, SELECT THE CORRECT LOAD PATTERN, FOR THE NUMBER OF UNIT LOADS TO BE SHIPPED, FROM TABLE I.
2. POSITION UNIT LOADS ON THE TRAILER AS SHOWN IN TABLE I. BLOCK AND TIE DOWN USING THE PRINCIPLES SHOWN ON PAGES 3 AND 4 (ALSO PAGE 6 WHEN VAN IS PERMITTED) AND THE GENERAL DOCUMENT MIL-STD-1320 (NAVY).

TABLE I

LOAD PATTERN *



* LOAD PLAN SHOWS FLOOR PLAN FOR TRAILER.
ARROWS INDICATE NOSE END OF BOMBS.
** 40 FT THROUGH 48 FT TRAILERS
*** 48 FT TRAILERS ONLY

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MIL-STD-1320-53C (NAVY)

FTL 40 FT TRAILER (VAN TYPE) (STD VAN AND RAGTOP VAN)

WARNING

VANS ARE AUTHORIZED ONLY WHEN FLATBED TRAILERS ARE NOT AVAILABLE AND SHIPMENT MUST BE MADE BECAUSE OF MILITARY NECESSITY.

1. THIS LOAD PLAN IS BASED ON THE USE OF STANDARD VANS AND "RAG TOP VANS". ALL TRAILERS MUST HAVE AXLES LOCATED IN THE "WESTERN" POSITION (AT EXTREME REAR OF TRAILER). DO NOT USE TRAILERS WITH METAL FLOORS.
2. WHEN LOADING IN STANDARD VANS, ALWAYS USE TRAILERS DESIGNED FOR HEAVY LOADING. TRAILER CROSS MEMBERS SHOULD BE ON 8-INCH CENTERS MINIMUM. DO NOT USE HANDLING EQUIPMENT THAT IS TOO HEAVY FOR THE JOB. USE THE LIGHTEST HANDLING EQUIPMENT CONSISTENT WITH APPROVED SAFETY PRACTICES.
3. UNIT LOADS MAY BE LOADED IN "RAG TOP" VANS USING APPROPRIATE HOISTING EQUIPMENT.
4. SIDE CLEATS, ITEM 9, ARE LOCATED AGAINST THE TRAILER SIDE WALLS. THEY SHOULD BEAR AGAINST ALL THREE RUNNERS AND SHOULD NOT EXTEND BEYOND OUTBOARD RUNNERS.
5. CENTER CLEATS, ITEM 10, ARE LOCATED BETWEEN UNIT LOADS AS SHOWN AND MAY BE ONE PIECE, WIDTH TO SUIT, IF DESIRED. CENTER CLEATS SHOULD BEAR AGAINST ALL THREE ADAPTER RUNNERS AND SHOULD NOT EXTEND BEYOND OUTBOARD RUNNERS.

TRUCKLOAD DATA

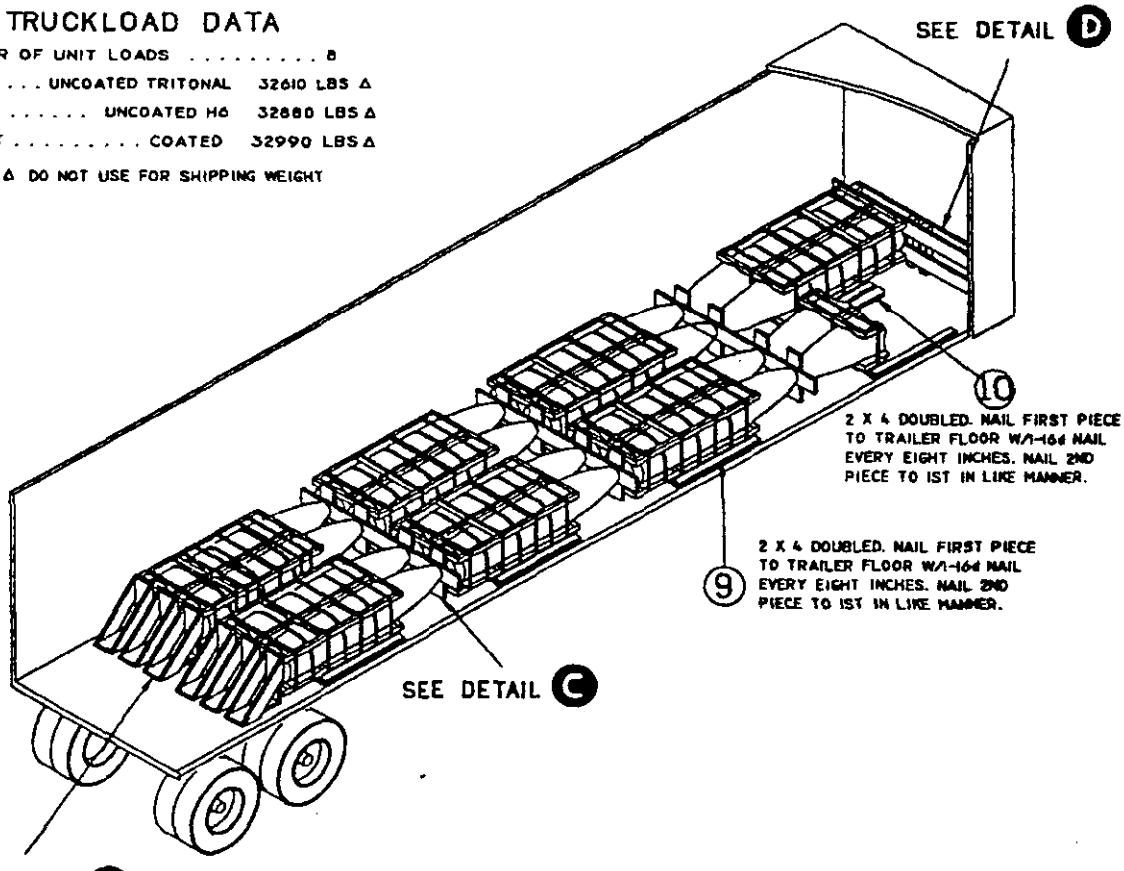
NUMBER OF UNIT LOADS 8

WEIGHT . . . UNCOATED TRITONAL 32610 LBS Δ

WEIGHT . . . . . UNCOATED H6 32880 LBS Δ

WEIGHT . . . . . COATED 32990 LBS Δ

Δ DO NOT USE FOR SHIPPING WEIGHT



WITH BRACE IN POSITION, NAIL ITEM 3 TO TRAILER FLOOR W/8-16d NAILS PER BRACE.

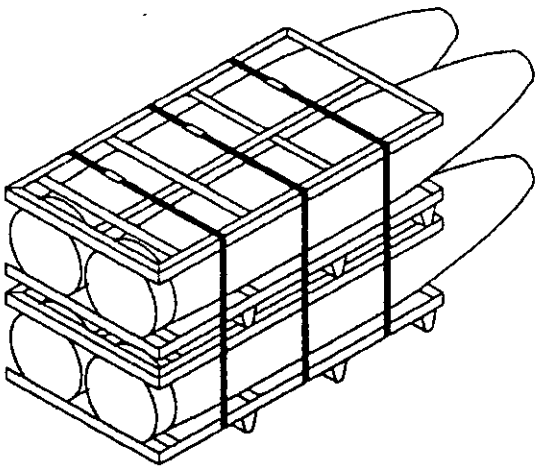
•TW EQUALS TRAILER WIDTH

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
16	10	CENTER CLEAT	SEE NOTE 11 ON PAGE 2	2 X 4 X 46
16	9	SIDE CLEAT	SEE NOTE 11 ON PAGE 2	2 X 4 X 46
2	8	FWD CROSS MEMBER	SEE NOTE 11 ON PAGE 2	2 X 8 X TO SUIT
2	7	AFT CROSS MEMBER	SEE NOTE 11 ON PAGE 2	2 X 8 X •TW-1/2
4	6	VERTICAL	SEE NOTE 11 ON PAGE 2	4 X 4 X 24
12	5	BRACE, DIAGONAL	SEE NOTE 11 ON PAGE 2	2 X 8 X TO SUIT
6	4	BRACE UPRIGHT	SEE NOTE 11 ON PAGE 2	2 X 8 X 24
6	3	BRACE, CLEAT	SEE NOTE 11 ON PAGE 2	2 X 8 X 22 1/2
12	2	VERTICAL	SEE NOTE 11 ON PAGE 2	2 X 8 X 24
3	1	HORIZONTAL	SEE NOTE 11 ON PAGE 2	2 X 8 X •TW-1/2

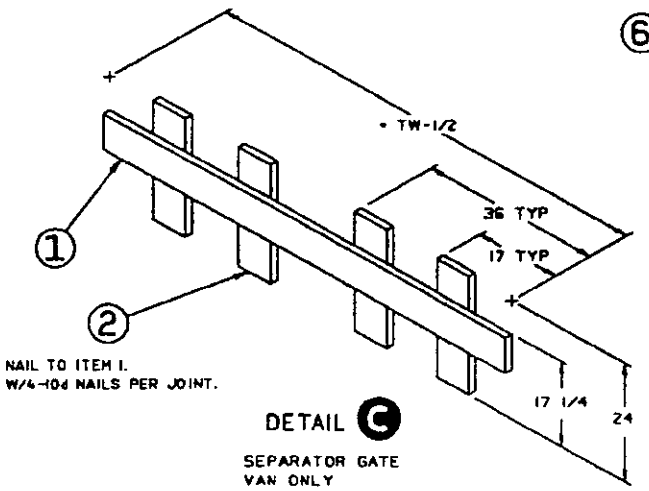
LIST OF MATERIALS

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UNITIZING DETAIL

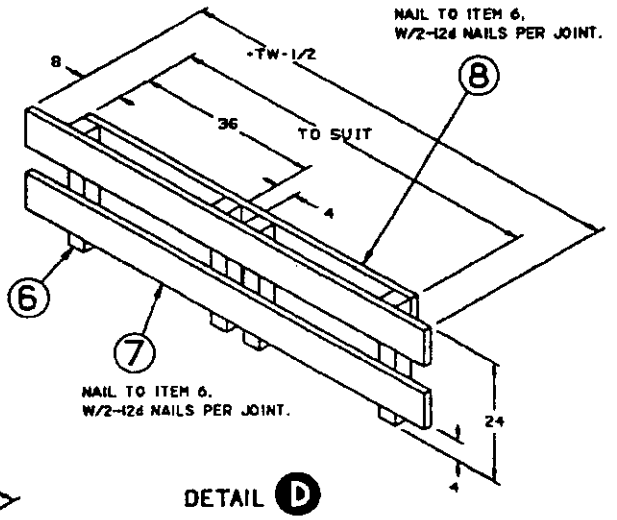


1. UNIT LOADS ARE STACKED AND STRAPPED IN PLACE ON THE TRAILER WITH LOADS IN EXACT POSITION.
2. STACK ONE UNIT LOAD ON TOP OF THE OTHER, WITH THE STACKING FEATURES ENGAGED.
3. THREAD STRAPS UNDER BOTTOM UNIT LOAD AND OVER TOP UNIT LOAD. TENSION AND DOUBLE NOTCH THE SEAL (SEE NOTE 11 ON PAGE 21).



DETAIL C

SEPARATOR GATE
VAN ONLY



DETAIL D

FRONT BUKHEAD
VAN ONLY

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